Michelle Thompson
CS340 module 7
Project Two
4/13/2026

README

This project is an interactive dashboard created for Grazioso Salvare. The main goal of the dashboard is to allow users to filter and view rescue dogs based on different rescue scenarios. The dashboard includes a data table that shows all animals, radio button filters for Water Rescue, Mountain or Wilderness Rescue, Disaster or Individual Tracking, and a Reset option, a pie chart that updates based on the selected data, and a map that displays location information. I also included the Grazioso Salvare logo and my name at the top. I tested each filter and took screenshots showing the dashboard working correctly for each scenario, including the reset returning all data, which proves the required functionality is complete.

To build this project, I used MongoDB, Python, Dash, Plotly, and Dash Leaflet. MongoDB was used to store and manage the animal data, and Python was used to connect to the database through a CRUD module. Dash was used to build the dashboard interface, while Plotly created the pie chart and Dash Leaflet handled the map. I used these tools because they all work well together and allowed me to build a full working dashboard using mostly Python instead of needing separate front-end languages.

MongoDB was used as the model component because it is a NoSQL database that stores data in a flexible format that works well with Python dictionaries. This made it easier to create and run queries using operators like $in, $gte, and $lte. MongoDB also handles large datasets and allows fast filtering, which is important for this type of application. From what I learned in the course,

MongoDB supports CRUD operations (create, read, update, delete), which are the main way data is managed and retrieved in this project .

The Dash framework acts as both the view and controller of the application. The layout I created is the view, which includes the table, filters, chart, and map. The controller part happens through callbacks. When a user clicks a filter, the callback runs a MongoDB query and updates everything on the screen at the same time. This helped me understand how user interaction connects directly to backend data.

Some resources I used during this project include https://dash.plotly.com/dash-core-components, https://dash.plotly.com/datatable, https://dash.plotly.com/datatable/interactivity, and https://jmeter.apache.org/. These helped me understand how to build the table, connect filters, and troubleshoot issues when my dashboard was not updating correctly.

To complete the project, I first imported the dataset into MongoDB and tested my CRUD Python module. Then I built the dashboard layout in Dash and added the data table. After that, I created the radio button filters and wrote the queries for each rescue type. Next, I connected everything using callbacks so the dashboard would update dynamically. I added the pie chart and map, tested each feature multiple times, fixed errors, and then captured screenshots of the final working dashboard.

I ran into several challenges during this project. One of the biggest issues was callback errors, especially duplicate callback outputs, which happened when I accidentally created more than one callback doing the same thing. That was confusing at first, but I fixed it by carefully checking my code and making sure each output only had one callback. I also struggled with knowing exactly where to place code in the template, and sometimes I put things in the wrong section

which caused errors. Another challenge was getting the filters to correctly match the MongoDB queries, especially making sure I used the right fields like breed, sex, and age ranges. The chart and map also did not show up at first, which made me think something was really wrong, but it ended up being how I passed the data into the callback. I had to slow down a lot during this project and go step by step, but that actually helped me understand everything better.

To reproduce this project, first install MongoDB and load the Austin Animal Center dataset into a database called aac. Then make sure Python and required libraries such as dash, plotly, pandas, and pymongo are installed. Next, run the CRUD Python module to confirm database connectivity. After that, open the ProjectTwoDashboard.ipynb file in Jupyter Notebook or Codio and run all cells. The dashboard will launch in the browser and allow interaction with the filters, table, chart, and map.

Overall, this project was challenging but also helped me understand how a real application works from front end to back end. I learned how to connect a database to a dashboard, how user input triggers changes, and how to troubleshoot errors when things do not work the first time. Even though I made mistakes along the way, I was able to fix them and complete a working dashboard that meets all the requirements.

GRAZIOSO SALVARE

Grazioso Salvare Dashboard

Michelle Thompson

○ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster or Individual Tracking ◉ Reset

	rec_num	age_upon_outcom	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	o
	filter data...										
◉	1	3 years	A746874	Cat	Domestic Shortha	Black/White	2014-04-10	2017-04-11 09:00	2017-04-11T09:00		SC
○	2	1 year	A725717	Cat	Domestic Shortha	Silver Tabby	2015-05-02	2016-05-06 10:49	2016-05-06T10:49		SC
○	4	7 months	A733653	Cat	Siamese Mix	Seal Point	2016-01-25	2016-08-27 18:11	2016-08-27T18:11	Kitty	
○	5	2 years	A691584	Dog	Labrador Retriev	Tan/White	2012-11-06	2015-05-30 13:48	2015-05-30T13:48	Luke	
○	7	2 years	A673830	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15	2014-03-19T15:15	*Seth	Ag
○	3	2 years	A716330	Dog	Chihuahua Shorth	Brown/White	2013-11-18	2015-12-28 18:43	2015-12-28T18:43	Frank	
○	6	5 years	A696004	Dog	Cardigan Welsh C	Sable/White	2010-01-27	2015-01-28 10:39	2015-01-28T10:39	Lucy	Ra
○	8	1 year	A736551	Dog	Labrador Retriev	Black	2015-10-12	2016-11-27 18:00	2016-11-27T18:00	*Mia	
○	9	3 years	A720214	Dog	Labrador Retriev	Red/White	2013-02-04	2016-02-11 12:41	2016-02-11T12:41	Blessing	
○	11	1 year	A721199	Dog	Dachshund Wireha	Tan/White	2015-02-23	2016-02-27 17:49	2016-02-27T17:49	Belle	

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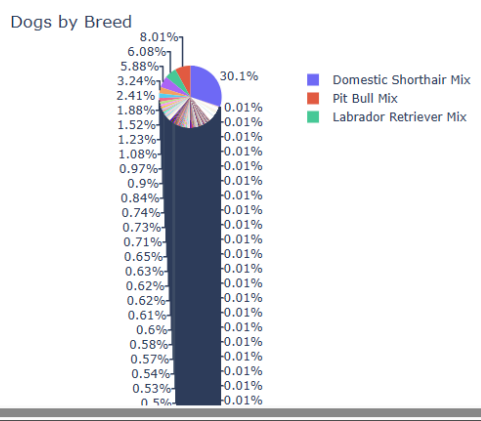

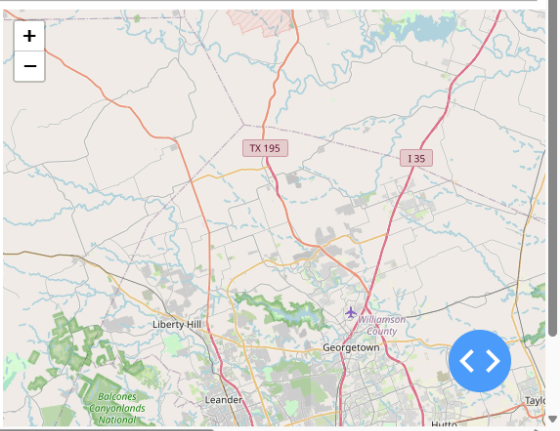

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◉ Water Rescue ○ Mountain or Wilderness Rescue ○ Disaster or Individual Tracking ○ Reset

	rec_num	age_upon_outcom	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	o
	filter data...										
◉	36	6 months	A706953	Dog	Labrador Retriev	Yellow	2014-12-06	2015-07-06 11:33	2015-07-06T11:33		Me
○	732	2 years	A749782	Dog	Labrador Retriev	Tan/White	2015-05-19	2017-07-25 14:59	2017-07-25T14:59	*Catalina	
○	1121	1 year	A757158	Dog	Labrador Retriev	White/Black	2016-08-30	2017-08-31 14:12	2017-08-31T14:12	Pirata	
○	1628	9 months	A740471	Dog	Labrador Retriev	Tan/White	2016-03-17	2016-12-23 17:13	2016-12-23T17:13	Mika	
○	1757	7 months	A742767	Dog	Labrador Retriev	Black	2016-06-27	2017-02-14 15:20	2017-02-14T15:20	Marley	
○	1988	1 year	A762781	Dog	Labrador Retriev	Black/White	2016-11-27	2017-12-03 13:09	2017-12-03T13:09		Pa
○	2041	2 years	A702745	Dog	Labrador Retriev	Black	2013-05-22	2015-05-22 11:45	2015-05-22T11:45	Abigail	
○	2225	2 years	A757341	Dog	Labrador Retriev	Black/White	2015-09-01	2017-10-03 12:27	2017-10-03T12:27	19	Pa
○	3319	9 months	A687748	Dog	Labrador Retriev	Yellow	2013-12-09	2014-09-09 17:01	2014-09-09T17:01		Su
○	4222	1 year	A735551	Dog	Labrador Retriev	Black	2015-09-25	2016-09-27 14:10	2016-09-27T14:10	Daisy	

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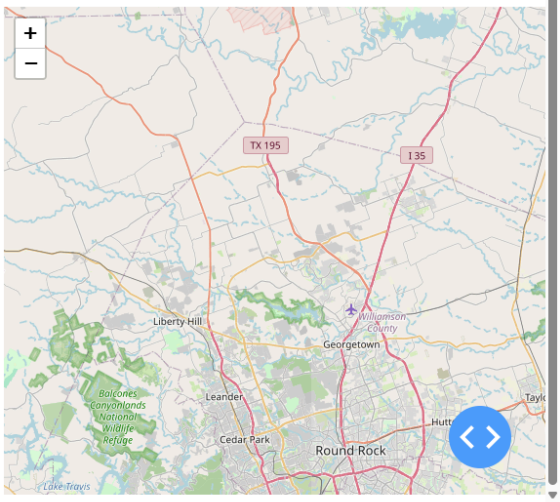
Dogs by Breed



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	rec_num	age_upon_outcom	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	o
	filter data...										
◉	5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24	2015-08-02T17:24	Papa	
○	5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24	2015-08-02T17:24	Papa	
○	5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24	2015-08-02T17:24	Papa	
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54	2018-01-22T11:54	Sargent	
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54	2018-01-22T11:54	Sargent	
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54	2018-01-22T11:54	Sargent	
○	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23	2017-09-23T11:23	Zeke	
○	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23	2017-09-23T11:23	Zeke	
○	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23	2017-09-23T11:23	Zeke	
○	3130	2 years	A721834	Dog	Siberian Husky	Brown/White	2014-03-05	2016-03-23 16:23	2016-03-23T16:23		Su

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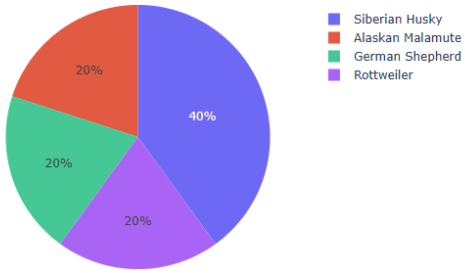
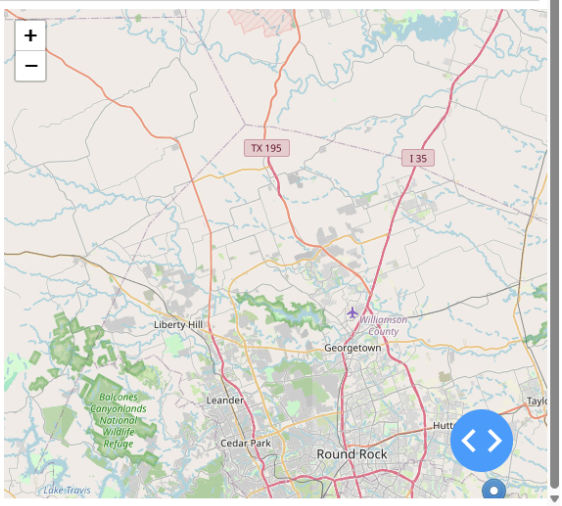
Dogs by Breed





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filter data...										
3767	4 years	A712291	Dog	Bloodhound	Red	2011-09-20	2015-09-22 15:43	2015-09-22T15:43	Boomer	
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2987	4 years	A694614	Dog	Rottweiler	Black/Brown	2011-01-01	2015-01-01 14:25	2015-01-01T14:25	Striker	
6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23	2017-09-23T11:23	Zeke	
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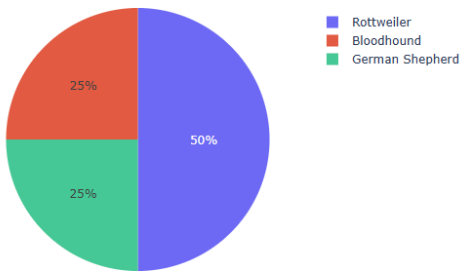

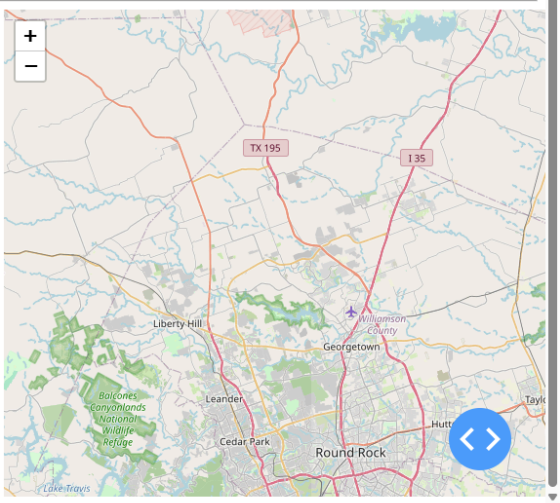

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○	6	5 years	A696004	Dog	Cardigan Welsh C	Sable/White	2010-01-27	2015-01-28 10:39	2015-01-28T10:39	Lucy	Ra
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○	9	3 years	A720214	Dog	Labrador Retriev	Red/White	2013-02-04	2016-02-11 12:41	2016-02-11T12:41	Blessing	
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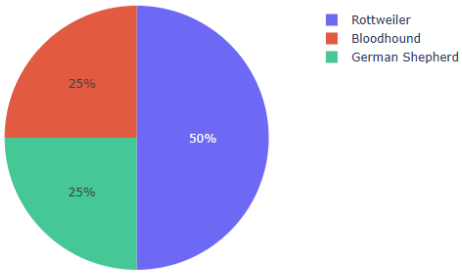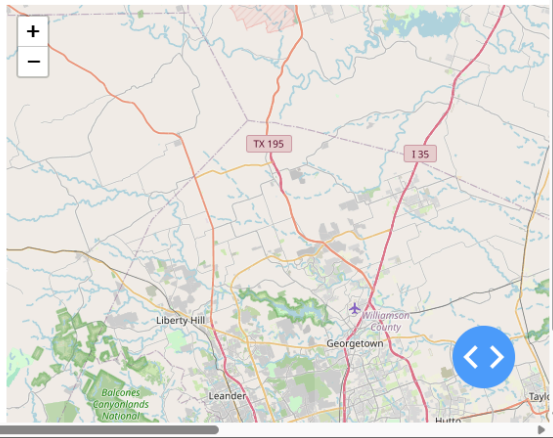
Dogs by Breed